

Ayoub Mnina

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SUMMARY

Computer Science Engineer graduated from INP-ISIMA, specializing in applied mathematics and data science. With nearly 2 years of experience in data pipelines, machine learning, and deployment of analytical solutions in industrial and space environments. Immediately available, open to relocation internationally (EU, Canada, USA, MENA).

EXPERIENCE

Scientific Software Engineer — Data & Analytics

Apr 2025 – Oct 2025

CNES – French Space Agency (*Intern*)

Toulouse, France

- Designed and implemented Python scripts and pipelines for the collection, transformation, and processing of large-scale data from the *PPP-WIZARD* computation chain (precise GNSS satellite orbit and clock estimation)
- Developed automated reporting and real-time monitoring solutions in Python to track system performance — delivering significant time savings for the internal team
- Built unsupervised learning models (clustering) for the analysis of large-scale datasets
- *Stack* : Python, Pandas, Scikit-learn, Plotly/Dash, Linux, GitLab, Confluence/Jira

Data Scientist/Engineer

Jun 2024 – Sep 2024

AW Service (*Intern*)

Paris, France

- Built an ETL pipeline for hotel booking data analysis and business insights extraction
- Developed and trained predictive ML models for booking cancellation forecasting (89% precision), deployed in production
- *Stack* : Python, Scikit-learn, SQL/Snowflake, Qt, Power BI, Git, Azure

Scientific Software Engineer

Apr 2023 – Aug 2023

Dufournier Technologies (*Intern*)

Clermont-Ferrand, France

- Built an internal software automating the full data processing workflow for raw tire testing machine data (200 Hz time series): collection, cleaning, mathematical modelling, scientific computation of tyre performance metrics, storage in database (200+ tires), and advanced analysis and visualisation
- *Stack* : Python, NumPy/SciPy, PyQt5, PyQtGraph, SQL, SQL Server, Azure, Matlab

EDUCATION

MSc. in Computer Science — Applied Mathematics & Data Science

2021 – 2025

INP-ISIMA (*Institut Supérieur d'Informatique, de Modélisation et de leurs Applications*)

Clermont-Ferrand, France

International Exchange Program

2022

OTH Regensburg

Regensburg, Germany

- Relevant courses: Online Optimization, Operations Research, German language

Classes préparatoires, Mathematics, Physics, Computer Science

2019 – 2021

CPGE Lycée Moulay Idriss

Fez, Morocco

- Two-year post secondary program in advanced mathematics and physics

PROJECTS

- **Better Call SpaceLLM — AI Agent for International Space Law** 2026 (ongoing)
Built a multilingual RAG agent on a space law legal corpus (Python, LangChain, Llama, Streamlit)
- **Deep Learning — Image Classification (Mushroom Challenge)** 2025 (80h)
Developed CNN models with data augmentation and fine-tuning for image classification (PyTorch)
- **Sentiment Analysis & Customer Behaviour Prediction — British Airways** 2024 (45h)
Web scraping, NLP sentiment analysis and ML models to predict ticket purchasing behaviour (NLTK, Scikit-learn)
- **Volcano Detection on Mars — Satellite Image Classification** 2023 (60h)
Trained CNN/ML models for volcanic structure detection on Mars satellite imagery (Scikit-learn, OpenCV, TensorFlow)

SKILLS

Programming	Python, SQL, PySpark, R, Matlab, C/C++, Java
Data Science	Pandas, NumPy/SciPy, Scikit-learn, TensorFlow/Keras, PyTorch, OpenCV, NLTK, LangChain
Data Engineering	Databricks, Snowflake, Azure Data Factory, Microsoft Fabric, SQL Server, NoSQL, Talend
BI & Visualization	Power BI, Tableau, Plotly/Dash, Streamlit
Tools	Linux, Git/GitLab, Docker, Jupyter, QT, FastAPI, CPLEX/GurobiPy, MPI/OpenMP, LaTeX
Languages	French (Proficient), English (Proficient — TOEIC 920), Arabic (native), German (beginner)